

Web Content Accessibility Guidelines (WCAG) 2.2 (<https://www.w3.org/TR/WCAG22/>)

12 December 2024

Conformance levels (A-AAA, low-high)

The Five Requirements to be classified as ‘conforming’ to WCAG2

1. All criteria of a certain level of conformance is met in full (if part of a higher compliance level is complete, it still remains at the highest fully-met level)
2. Every part of the site must be included and conform to the standard- it cannot be achieved if part of the webpage is excluded
3. If a process includes several web pages (steps that take multiple pages to complete a task, such as adding items to cart, inputting payment and shipping information, and checking out), all web pages involved in that process must be at the same level (or higher)
4. Criteria must be met using accessibility-supported ways of using technologies, and any information that is presented in a non-conforming manner must also be available in an accessibility supported manner
5. If there are elements that are not conforming, they must not block access from the rest of the page.

The Criteria (and the conformance level)

1. Percieveable

1.1. Text alternatives -Provide text alternatives for any non-text content

- 1.1.1. All non-text content that is presented to the user has a text alternative that serves the equivalent purpose, except for the situations listed below.

Controls, Input: If it is controlled or accepts user input, the content needs a name that indicates what it controls/requires

1.1.1.1.1 Text fields are to have names, buttons are to be labelled

Test: If the purpose of the non-text content would be invalid if it had been presented in text, text alternatives would provide descriptive identification of the content

1.1.1.1.2 An image is shown and is to be identified as a test. Instead of the alternative text calling the object by the name to be identified, it would describe what the image shows

Sensory: If the purpose of the non-text information is for sensory experience and does not primarily convey important information or perform a function, descriptive identification is to be provided

1.1.1.1.3 Musical performance, visual art

CAPTCHA: If the non-text information is used as a CAPTCHA, descriptive identification should be provided and alternate forms of CAPTCHA using different output modes should be available to accommodate different disabilities

1.1.1.1.4 Image selection, distorted character reading, sound recognition, logic prompt

Decoration, Formatting, Invisible: If the non-text information is PURELY DECORATIONAL or not visible to the user,, it must be implemented in a way that it is ignored by assistive technologies like screen readers (A)

1.2. “Time-based Media” - Provide alternatives for time-based media.

1.2.1. Prerecorded Audio-Only or Video-only (A)

1.2.1.1. A non-time based alternate is provided that presents equivalent information for pre-recorded audio-only content

1.2.1.1.1 A transcript is available as an alternative to an audio recording

1.2.1.2. Either an alternate for time-based media or an audio track with equivalent information is to be provided with pre-recorded video-only content

1.2.2. Prerecorded audio content must have captions that are synchronized with the video **(A)**

1.2.3. If synchronized media includes video content, an alternative for the time-based media or an audio description must be available **(A)**

1.2.4. In synchronized media with live audio content, captions are to be provided **(AA)**

1.2.5. **(AAA)**

1.2.6. **(AAA)**

1.2.7. **(AAA)**

1.3. Adaptable: Create content that can be presented in different ways (for example simpler layout) without losing information or structure.

1.3.1. Information, structure, and relationships conveyed through presentation can be programmatically determined or are available in text. **(A)**

1.3.1.1. Allowing fast and easy access to information by code in the same way that visual navigation allows. For example, required text fields are marked with asterisks.

- 1.3.1.2. Assistive technology needs to be able to tell what different relevant elements are.
- 1.3.2. When the sequence in which content is presented affects its meaning, a correct reading sequence can be programmatically determined. **(A)**
 - 1.3.2.1. Use code to preserve meaningful content order
 - 1.3.2.2. Only one correct order needs to be provided
- 1.3.3. Instructions provided for understanding and operating content do not rely solely on sensory characteristics of components such as shape, color, size, visual location, orientation, or sound. **(A)**
 - 1.3.3.1. Describe controls by name, not just by appearance or location.
- 1.3.4. Content does not restrict its view and operation to a single display orientation, such as portrait or landscape, unless a specific display orientation is essential.
(AA)
 - 1.3.4.1. Don't lock content to either portrait or landscape mode
- 1.3.5. The purpose of each input field collecting information about the user can be programmatically determined when:
 - The input field serves a purpose identified in the Input Purposes for user interface components section; and
 - The content is implemented using technologies with support for identifying the expected meaning for form input data. **(AA)**
 - 1.3.5.1. Use code to indicate the purpose of common inputs, where technology allows.

1.3.5.2. Clarify the desired data and purpose of the information for forms, especially for autocomplete

1.3.6. (AAA)

1.4. Distinguishable: Make it easier for users to see and hear content including separating foreground from background.

1.4.1. Color cannot be the only visual means of conveying information, prompting response, or distinguishing visual elements (A)

1.4.1.1. To meet this, add additional forms of visual distinction- graphs get shape/line differentiation, links get underlined, etc...

1.4.2. If any audio automatically plays on a web page (more than 3 seconds) there must be a method to pause/stop the audio available, or the user must be able to control the audio volume independent from the overall system volume (A)

1.4.3. Visual presentation of text (including images of text) have a contrast ratio 4.5:1 or more, apart from the following:

Large Text: Large-scale text or images of large text must have ratio 3:1

Incidental Text: text that is part of inactive component, purely decorative, invisible, or part of image that contains other visual content does not have a contrast requirement

Logos: Text within logos and brand names have no contrast requirements

(AA)

1.4.4. Text must be resizable without assistive technology up to 200% without loss of content or functionality (AA)

1.4.4.1. Exceptions are captions and images of text

- 1.4.5. If it is possible to achieve the visual presentation in text, text is used instead of images of text apart from the exceptions:

Customizable: The image of text can be customized to the user's requirements

Essential: The specific presentation of text is essential to the information being conveyed (this includes logotypes) **(AA)**

- 1.4.6. **(AAA)**

- 1.4.7. **(AAA)**

- 1.4.8. **(AAA)**

- 1.4.9. **(AAA)**

- 1.4.10. Content can be presented without loss of information of functionality, and without scrolling in 2-dimensions for vertical scrolling content at a width equivalent to 320 CSS pixels or horizontal scrolling content at a height equivalent to 256 CSS pixels. **(AA)**

- 1.4.10.1. This is saying that when the page is resized, it is not required to scroll in both directions to view information

- 1.4.11. Visual contrast between adjacent colors must be at least 3:1 for the following:

User interface components, controls, and status information (Buttons, checkboxes, toggles, text boxes, etc)

Graphic objects unless particular presentation is vital to the information conveyed (Icons, graphs, charts, infographics)

(AA)

1.4.11.1. This has overlap with Focus Visible (2.4.7) as the added focus must also comply with this

1.4.12. Text format can be overridden from the original spacing values to the following specifications without losing content or functionality:

Line height (line spacing) to at least 1.5 times the font size;

Spacing following paragraphs to at least 2 times the font size;

Letter spacing (tracking) to at least 0.12 times the font size;

Word spacing to at least 0.16 times the font size.

(AA)

1.4.13. If receiving and then removing focus (pointer or keyboard) triggers additional content to become visible and then hidden, the following are true:

Dismissable: The additional content can be dismissed without removing focus from the object, unless it conveys an error or does not obscure/replace content

Hoverable: If activated by pointer hover, the pointer can be moved over the additional content without it disappearing

Persistent: The additional content does not disappear unless the focus has been removed, the user dismisses the content, or its information is no longer

valid **(AA)**

2. **Operable**

2.1. **Keyboard Accessible**

2.1.1. All functionality of the content is operable through a keyboard interface without requiring specific timings for individual keystrokes, except where the

underlying function requires input that depends on the path of the user's movement and not just the endpoints. **(A)**

2.1.1.1. All actions possible with a mouse can be done with purely keyboard input as well

2.1.2. If keyboard focus can be moved to a component of the page using a keyboard interface, then focus can be moved away from that component using only a keyboard interface, and, if it requires more than unmodified arrow or tab keys or other standard exit methods, the user is advised of the method for moving focus away. **(A)**

2.1.2.1. If any non-standard movements are required, the user is informed

2.1.3. **(AAA)**

2.1.4. If a keyboard shortcut is implemented in content using only letter (including upper- and lower-case letters), punctuation, number, or symbol characters, then at least one of the following is true:

Turn off: A mechanism is available to turn the shortcut off;

Remap: A mechanism is available to remap the shortcut to include one or more non-printable keyboard keys (e.g., Ctrl, Alt);

Active only on focus: The keyboard shortcut for a user interface component is only active when that component has focus. **(A)**

2.1.4.1. Character-only shortcut keys need to be able to be turned off or modifiable

2.2. Enough Time

2.2.1. For each time limit that is set by the content, at least one of the following is true:

Turn off: The user is allowed to turn off the time limit before encountering it

Adjust: The user is allowed to adjust the time limit before encountering it over a wide range that is at least ten times the length of the default setting;

Extend: The user is warned before time expires and given at least 20 seconds to extend the time limit with a simple action (for example, "press the space bar"), and the user is allowed to extend the time limit at least ten times

Real-time Exception: The time limit is a required part of a real-time event (for example, an auction), and no alternative to the time limit is possible

Essential Exception: The time limit is essential and extending it would invalidate the activity

20 Hour Exception: The time limit is longer than 20 hours. (A)

2.2.2. For moving, blinking, scrolling, or auto-updating information, all of the following are true:

For any moving, blinking or scrolling information that (1) starts automatically, (2) lasts more than five seconds, and (3) is presented in parallel with other content, there is a mechanism for the user to pause, stop, or hide it unless the movement, blinking, or scrolling is part of an activity where it is essential; and

For any auto-updating information that (1) starts automatically and (2) is presented in parallel with other content, there is a mechanism for the user to

pause, stop, or hide it or to control the frequency of the update unless the auto-updating is part of an activity where it is essential. **(A)**

2.2.3. **(AAA)**

2.2.4. **(AAA)**

2.2.5. **(AAA)**

2.2.6. **(AAA)**

2.3. Seizures and Physical Reactions

2.3.1. Web pages do not contain anything that flashes more than three times in any one second period, or the flash is below the general flash and red flash thresholds. **(A)**

2.3.1.1. There are no more than three general flashes and/or no more than three red flashes within any one-second period; or

2.3.1.2. The combined area of flashes occurring concurrently occupies no more than a total of .006 steradians within any 10 degree visual field on the screen (25% of any 10 degree visual field on the screen) at typical viewing distance

2.3.1.3. **A general flash** is defined as a pair of opposing changes in relative luminance of 10% or more of the maximum relative luminance (1.0) where the relative luminance of the darker image is below 0.80; and where "a pair of opposing changes" is an increase followed by a decrease, or a decrease followed by an increase, and

2.3.1.4. **A red flash** is defined as any pair of opposing transitions involving a saturated red

2.3.2. (AAA)

2.3.3. (AAA)

2.4. Navigable

2.4.1. A mechanism is available to bypass blocks of content that are repeated on multiple web pages. (A)

2.4.1.1. Provide a means of skipping repeating content.

2.4.2. Web pages have titles that describe their topic or purpose. (A)

2.4.2.1. Titles identify the current location without requiring users to read or interpret page content.

2.4.2.2. the title of the page should also be changed dynamically to reflect the content or topic of the current view.

2.4.3. If a web page can be navigated sequentially and the navigation sequences affect meaning or operation, focusable components receive focus in an order that preserves meaning and operability. (A)

2.4.3.1. Navigating with only the keyboard must make sense.

2.4.3.2. The index table specifically should be looked at to make sure the order makes sense

2.4.4. The purpose of each link can be determined from the link text alone or from the link text together with its programmatically determined link context, except where the purpose of the link would be ambiguous to users in general. (A)

2.4.4.1. Provide descriptive names or context for all links.

2.4.5. More than one way is available to locate a web page within a set of web pages except where the web page is the result of, or a step in, a process. **(AA)**

2.4.5.1. Provide at least two options for reaching the same content.

2.4.6. Headings and labels describe topic or purpose. **(AA)**

2.4.6.1. This success criterion requires that if headings or labels are provided, they be descriptive

2.4.7. Any keyboard operable user interface has a mode of operation where the keyboard focus indicator is visible. **(AA)**

2.4.7.1. Ensure each item receiving focus has a visible indicator.

2.4.7.2. Examples being highlighted text, visible border, etc

2.4.8. **(AAA)**

2.4.9. **(AAA)**

2.4.10. **(AAA)**

2.4.11. When a user interface component receives keyboard focus, the component is not entirely hidden due to author-created content. **(AA)**

2.4.11.1. Ensure when an item gets keyboard focus, it is at least partially visible.

2.4.11.2. The component receiving focus should never be obscured to the point a user cannot tell which item has focus.

2.4.12. **(AAA)**

2.4.13. **(AAA)**

2.5. Input Modalities

2.5.1. Functions that use multipoint or path-based gestures have alternate methods of operation unless it is essential **(A)**

- 2.5.1.1. Pinch to zoom has buttons that can perform same action, slideshow that gets advanced with a swipe also has a button, a slider also has increase/decrease buttons
- 2.5.2. For single-pointer functions, one of the following must be true-
 - The down-event of the pointer does not execute any part of the function
 - The up-event is required to complete the function, and there are ways to abort the function before completion or undo after completion
 - The up-event reverses the outcome of the preceding down-event or
 - The function getting completed on the down-event is essential **(A)**
- 2.5.2.1. This ensures that accidental actions can be undone or aborted easily
- 2.5.3. For user interface components with labels that include text or images of text, the name contains the text that is presented visually. **(A)**
 - 2.5.3.1. Make the control's text label and name match.
 - 2.5.3.2. If the label uses symbolic text characters, the accessible name should be the function the button serves ("B" would have the name "Bold" when used for that function)
 - 2.5.3.2.1 This does not apply when using symbols in mathematical expressions/formulae, as it should be exactly the same as the text (and the proper symbols should be utilized)
 - 2.5.3.3. If the label contains parenthetical information it may not be needed as part of the name unless it gives important context
 - 2.5.3.3.1 Creating an accessible name for a field with the label "Name (Required)" should include that it is required

- 2.5.4. Motion controlled (device or user motion) functions can be performed with user interface components and the motion controls can be disabled except when: **(A)**

The motion is used to operate functionality through an accessibility supported interface

The motion is essential for the function

- 2.5.4.1. This is to prevent accidental activation and unintentional changes

- 2.5.5. **(AAA)**

- 2.5.6. **(AAA)**

- 2.5.7. Functions that use dragging movements can be achieved with a single pointer without dragging, unless essential or user-defined **(AA)**

- 2.5.7.1. This is separate from path-based gestures (2.5.1) as the path taken does not matter for drag movements, so complying with 2.5.1 does not automatically comply with 2.5.7

- 2.5.7.2. Drag for color has option to type in a hex code instead, or ability to just click on the desired color on the wheel without dragging

- 2.5.7.3. Drag for navigation of map also has buttons to move the view

- 2.5.8. The size of the target for pointer inputs is at least 24 by 24 CSS pixels except for the following: **(AA)**

Targets that are too small are spaced so that there are no other targets (or other circles for undersized targets) within a 24 CSS pixel diameter circle placed at the center of the smallest rectangle in which all the points of the target lie

The function of the button can be performed with a different control on the

same page that meets this criterion

The target is within a sentence or its size is similarly constrained by the line-height of non-target text

The size of the target is determined by the user alone

The size of the target is essential or legally required for the information being conveyed

3. Understandable

3.1. Readable

3.1.1. The default human language of each web page can be programmatically determined. **(A)**

3.1.1.1. Indicate the predominant language on a page.

3.1.1.2. Use the lang attribute on the html element.

3.1.2. The human language of each passage or phrase in the content can be programmatically determined except for proper names, technical terms, words of indeterminate language, and words or phrases that have become part of the vernacular of the immediately surrounding text. **(AA)**

3.1.2.1. Indicate when words are in a different language.

3.1.3. **(AAA)**

3.1.4. **(AAA)**

3.1.5. **(AAA)**

3.1.6. **(AAA)**

3.2. Predictable

3.2.1. When a user interface component receives focus, it does not initiate a change of context **(A)**

3.2.1.1. Examples include (not limited to):

forms submitted automatically when a component receives focus;

new windows launched when a component receives focus;

focus is changed to another component when that component receives focus;

3.2.2. Changing the setting of any user interface component does not automatically cause a change of context unless the user has been advised of the behavior before using the component. **(A)**

3.2.2.1. Changing the setting of a user interface component is changing some aspect in the control that will persist when the user is no longer interacting with it. So checking a checkbox, entering text into a text field, or changing the selected option in a list control changes its setting, but activating a link or a button does not

3.2.3. Navigation menus that repeat on multiple web pages within a set of pages should always be in the same order, unless the user initiates a change **(AA)**

3.2.4. Components that have the same function within a set of web pages are identified consistently **(AA)**

3.2.5. **(AAA)**

3.2.6. If a web page contains any of the following help mechanisms and the mechanisms are repeated across multiple pages, they are in the same order relative to other content (unless the user initiates the change) **(A)**

Human contact details;

Human contact mechanism;

Self-help option;

A fully automated contact mechanism.

3.3. Input Assistance

3.3.1. If an input error is automatically detected, the item that is in error is identified and the error is described to the user in text. **(A)**

3.3.1.1. Provide descriptive notification of errors.

3.3.1.2. If a user enters a value that is too high or too low, and the coding on the page automatically changes that value to fall within the allowed range, the user's error would still need to be described to them as required by the success criterion.

3.3.1.3. It is perfectly acceptable to indicate the error in other ways such as through the use of an image, color, or other visual indicator, *in addition to the text description*.

3.3.2. Labels or instructions are provided when content requires user input. **(A)**

3.3.2.1. Provide labels or instructions for inputs.

3.3.2.2. Too much information or instruction can be just as harmful as too little.

3.3.3. If an input error is automatically detected and suggestions for correction are known, then the suggestions are provided to the user, unless it would jeopardize the security or purpose of the content. **(AA)**

- 3.3.3.1. The definition of "input error" says that it is "information provided by the user that is not accepted" by the system.
- 3.3.3.2. Where errors are detected, suggest known ways to correct them.
- 3.3.4. For web pages that cause legal commitments or financial transactions for the user to occur, that modify or delete user-controllable data in data storage systems, or that submit user test responses, at least one of the following is true:
- Submissions are reversible.
 - Data entered by the user is checked for input errors and the user is provided an opportunity to correct them.
 - A mechanism is available for reviewing, confirming, and correcting information before finalizing the submission. **(AA)**
- 3.3.4.1. User-controllable data is user-viewable data that the user can change and/or delete through an intentional action. Examples of the user controlling such data would be updating the phone number and address for the user's account, or deleting a record of past invoices from a website. It does not refer such things as internet logs and search engine monitoring data that the user can't view or interact with directly.
- 3.3.5. **(AAA)**
- 3.3.6. **(AAA)**
- 3.3.7. Information previously entered by or provided to the user that is required to be entered again in the same process is either:
- auto-populated, or*

available for the user to select.

Except when:

re-entering the information is essential,

the information is required to ensure the security of the content, or

previously entered information is no longer valid. (A)

- 3.3.7.1. Don't ask for the same information twice in the same activity unless necessary.
- 3.3.7.2. The autocomplete feature of browsers is not considered sufficient because it is the content (the website) that needs to provide the stored information for a redundant entry, or avoid asking for the same information again.
- 3.3.7.3. This success criterion does not apply if data is provided by the user with a different method, such as uploading a resume in a document format.
- 3.3.8. A cognitive function test (such as remembering a password or solving a puzzle) is not required for any step in an authentication process unless that step provides at least one of the following:

Alternative

Another authentication method that does not rely on a cognitive function test.

Mechanism

A mechanism is available to assist the user in completing the cognitive function test.

Object Recognition

The cognitive function test is to recognize objects.

Personal Content

The cognitive function test is to identify non-text content the user provided to the website. (AA)

- 3.3.8.1. "Object recognition" and "Personal content" may be represented by images, video, or audio.
- 3.3.8.2. Examples of mechanisms that satisfy this criterion include: support for password entry by password managers to reduce memory need, and copy and paste to reduce the cognitive burden of re-typing.
- 3.3.8.3. Don't make people solve, recall, or transcribe something to log in.
- 3.3.8.4. The Google sign in should be sufficient based on [this page](#)

3.3.9. (AAA)

4. Robust

4.1. Compatible

- 4.1.1. Obsolete and removed
- 4.1.2. For all user interface components (including but not limited to: form elements, links and components generated by scripts), the name and role can be programmatically determined; states, properties, and values that can be set by the user can be programmatically set; and notification of changes to these items is available to user agents, including assistive technologies.
 - 4.1.2.1. standard HTML controls already meet this success criterion when used according to specification

- 4.1.3. In content implemented using markup languages, status messages can be programmatically determined through role or properties such that they can be presented to the user by assistive technologies without receiving focus.